🛡️ 3: Group Policies

✅ Tasks

1.

Open Group Policy Management Console (GPMC) on your Windows Server and create a new GPO named 'PasswordPolicyDemo' that enforces a minimum password length of 8 characters. Link it to your test OU and verify the setting is applied using gpresult.

📝 **Answer:**

Steps:

1. Open **Server Manager → Tools → Group Policy Management**.
2. Expand your domain.
3. Right-click your test OU and select **Create a GPO in this domain, and Link it here**.
4. Enter the name `PasswordPolicyDemo` and click OK.
5. Right-click the GPO and select **Edit**.
6. Go to:

   ```
   Computer Configuration → Policies → Windows Settings → Security Settings → Account Policies → Password Policy
   ```

7. Double-click **Minimum password length**.
8. Select **Define this policy setting** and enter `8`.
9. Click **Apply → OK**.
10. On the client PC, open Command Prompt as Administrator and run:

```
gpupdate /force
gpresult /r
```

11. Check the applied Group Policy information.

⚠️ **Note:** In a domain, password policies for domain accounts are normally configured in a GPO linked to the domain root. Linking this setting only to a test OU may not enforce it for domain users. Verify the effective domain password policy on the Domain Controller.

🎯 **Result:** The minimum password length was configured to 8 characters in the GPO.

2.

Configure a GPO to restrict access to the Control Panel for all users in a specific OU, similar to how public computers in a cybercafe or college lab prevent users from changing system settings.

Hint: In the GPO, navigate to User Configuration > Administrative Templates > Control Panel and set 'Prohibit access to Control Panel and PC settings' to Enabled.

📝 **Answer:**

Steps:

1. Open **Group Policy Management**.
2. Right-click the required OU and create a new GPO named `ControlPanelRestriction`.
3. Right-click the GPO and select **Edit**.
4. Go to:

   `User Configuration → Policies → Administrative Templates → Control Panel`

5. Double-click **Prohibit access to Control Panel and PC settings**.
6. Select **Enabled**.
7. Click **Apply → OK**.
8. Log in to the client PC using a user account from the target OU.
9. Run the following command:

```
gpupdate /force
```

10. Sign out and sign in again if needed, then try opening Control Panel.

🎯 **Result:** Users affected by this GPO are prevented from opening Control Panel and PC settings.

3.

Edit your 'PasswordPolicyDemo' GPO to enable account lockout after 3 invalid login attempts and set the lockout duration to 10 minutes. Run gpupdate /force on a client PC and test the lockout by entering the wrong password.

📝 **Answer:**

Steps:

1. Open **Group Policy Management**.
2. Right-click `PasswordPolicyDemo` and select **Edit**.
3. Navigate to:

```
Computer Configuration → Policies → Windows Settings → Security
Settings → Account Policies → Account Lockout Policy
```

4. Double-click **Account lockout threshold**.
5. Enable the setting and enter `3` invalid login attempts.
6. Click **Apply → OK**.
7. Double-click **Account lockout duration**.
8. Enter `10` minutes and click **Apply → OK**.
9. Also configure **Reset account lockout counter after** to `10` minutes.
10. On the client PC, open Command Prompt as Administrator and run:

```
gpupdate /force
```

11. Test using a dedicated test account by entering the wrong password three times.
12. Verify that the account becomes locked, then wait for the lockout period to expire or have an administrator unlock it.

⚠️ **Important:** For domain user accounts, account lockout policies are normally configured in a GPO linked at the domain level. A GPO linked only to a test OU may not control domain-account lockouts. Do not test with your main Administrator account.

🎯 **Result:** The account lockout policy was configured for three failed attempts and a 10-minute lockout duration.

4.

Document all GPO settings you applied in this session (password policy, desktop restriction, lockout policy) in a table format, and for each setting, describe its expected impact on users in 1-2 sentences.

📝 **Answer:**

GPO Setting	Configuration	Expected Impact on Users
Minimum Password Length	8 characters	Users must create passwords containing at least 8 characters when the policy applies.
Control Panel Restriction	Enabled	Targeted users cannot open Control Panel or PC settings to change system settings.
Account Lockout Threshold	3 invalid attempts	The account is locked after three failed login attempts when the policy applies.
Account Lockout Duration	10 minutes	A locked account remains locked for 10 minutes before it can automatically unlock.
Reset Lockout Counter	10 minutes	The failed-attempt counter resets after 10 minutes without another failed attempt.

🎯 **Result:** The GPO settings and their expected impact on users have been documented.

📚 Theory: What is Group Policy?

Group Policy is a Windows feature that allows administrators to manage and control user and computer settings centrally.

GPO (Group Policy Object): A collection of settings used to apply rules and restrictions to users or computers.

Uses of Group Policy:

- 🔐 Enforce password and account security policies.
- 🖥️ Restrict access to Control Panel.
- 🛡️ Apply security settings across computers.
- 🏢 Manage users and computers centrally.

🎉 Conclusion

In this practical, I created and configured Group Policy Objects for password length, Control Panel restrictions, and account lockout. I learned how to link GPOs, update policies using `gpupdate /force`, and verify applied policies using `gpresult`.

These settings help administrators improve security and control user access in an organization.